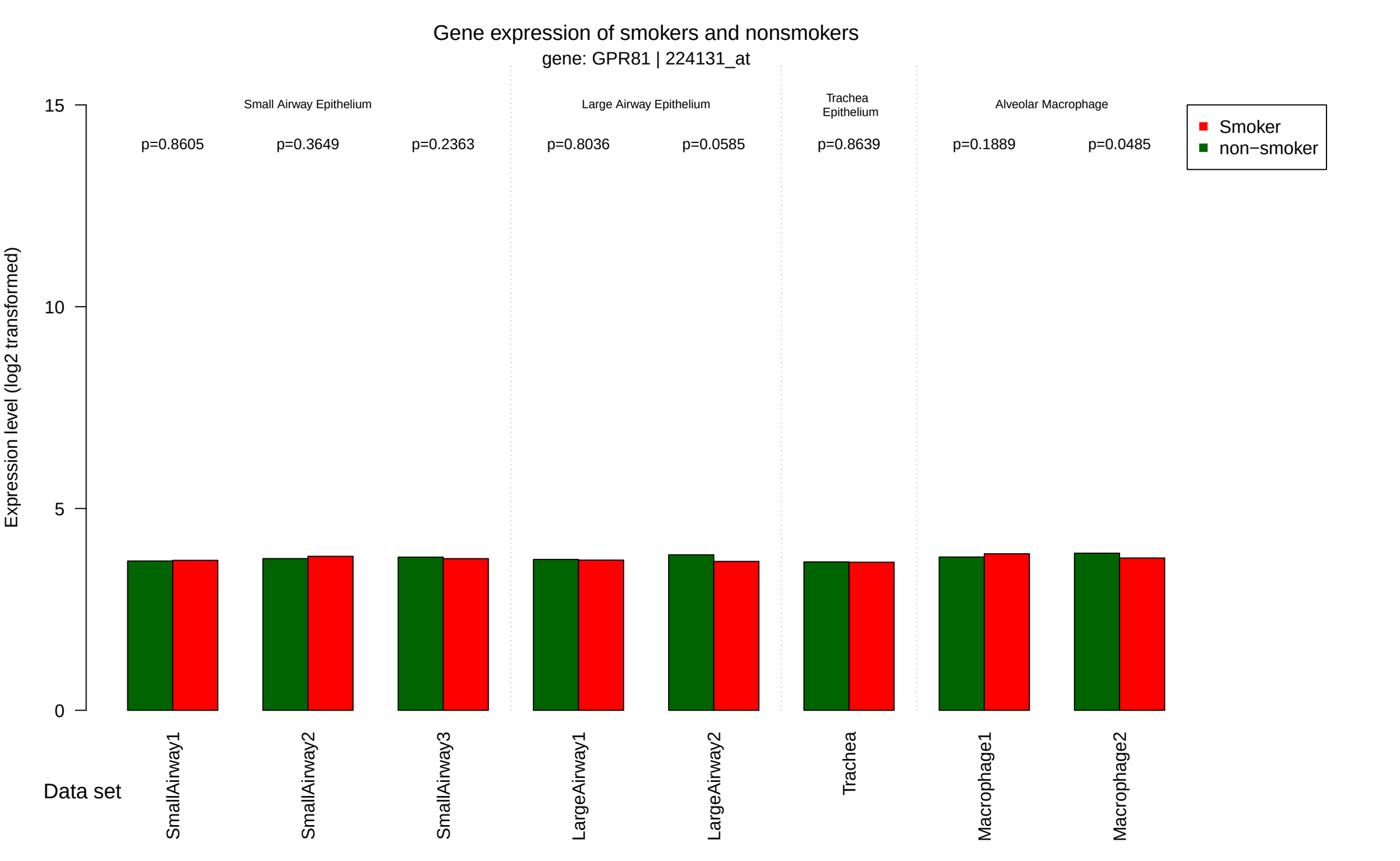
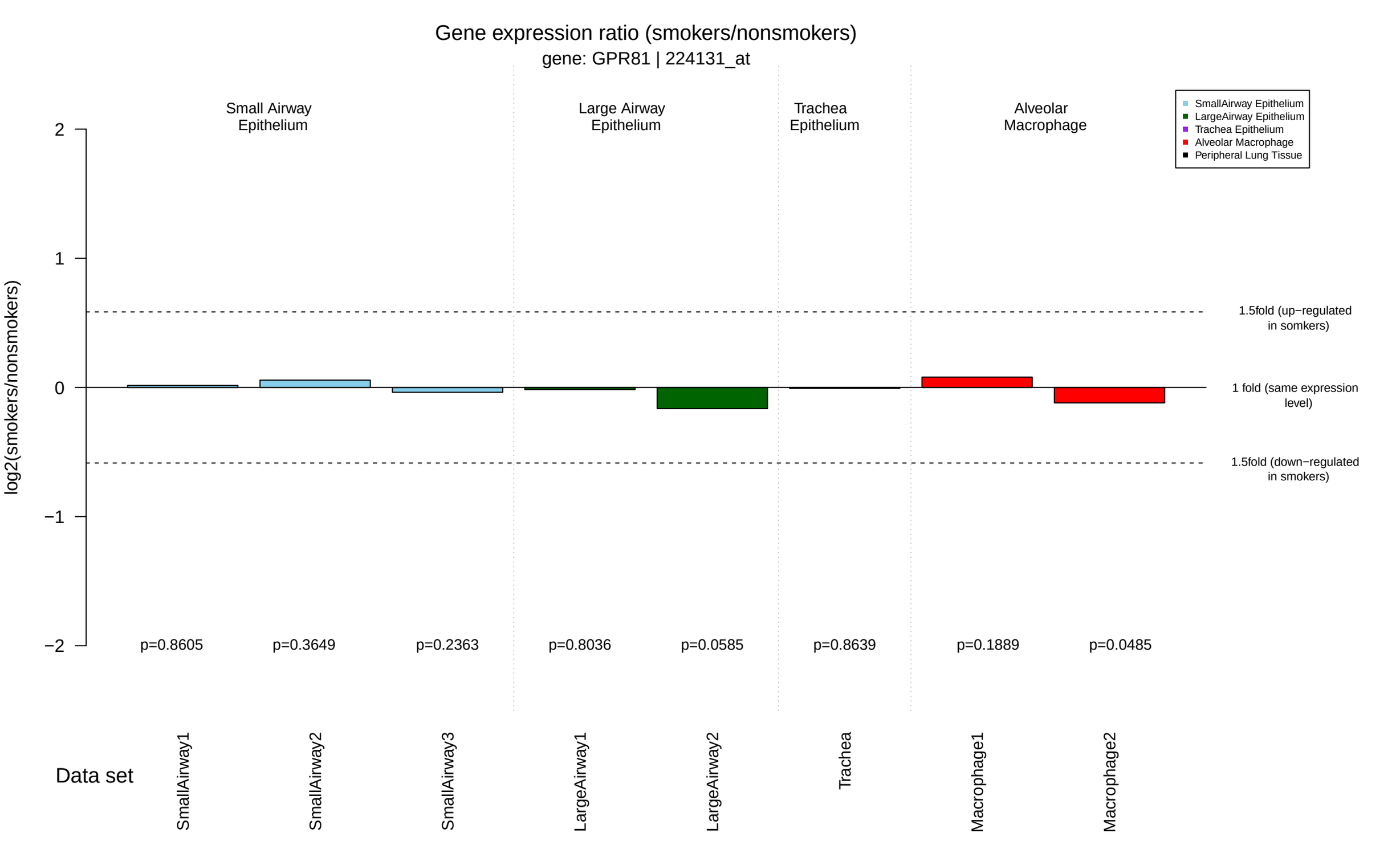
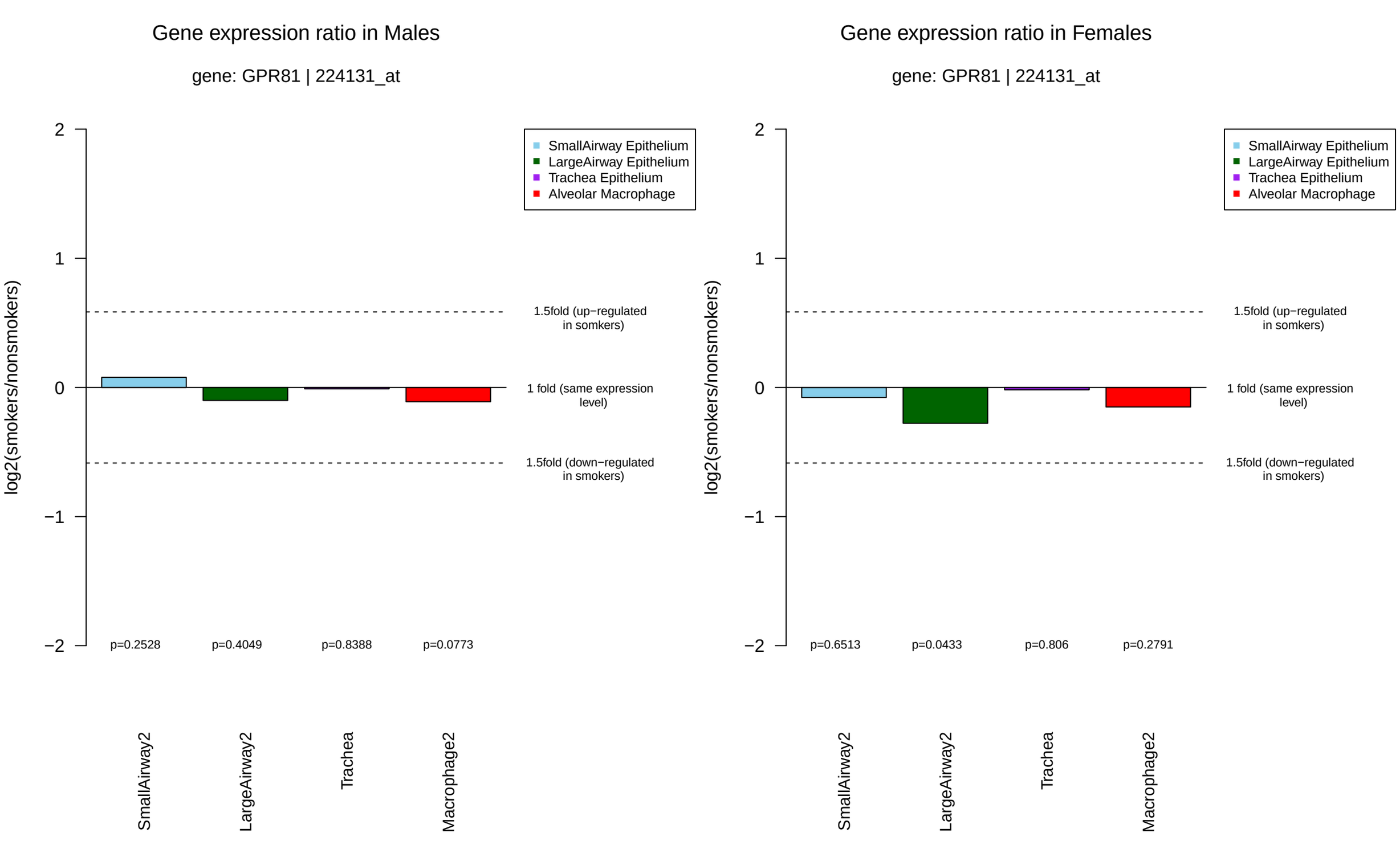
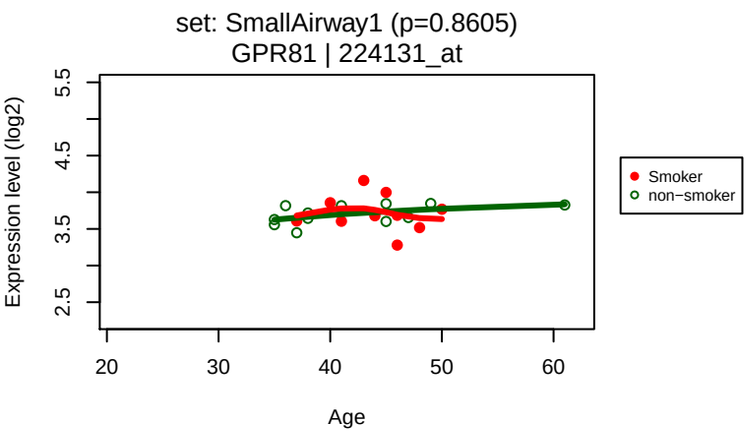
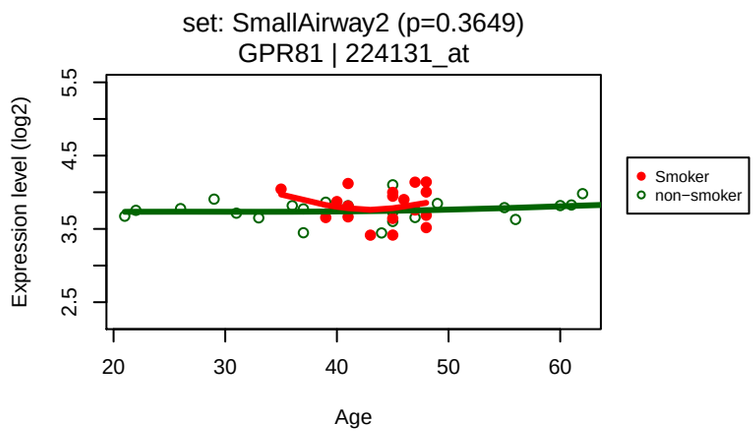
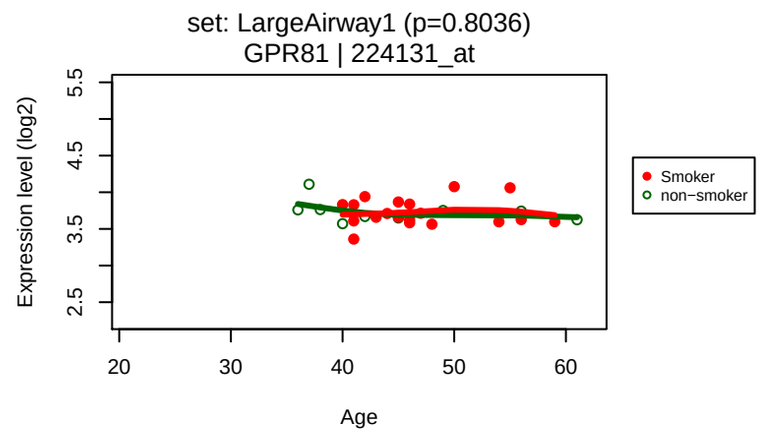

Gene expression of smokers and nonsmokers
gene: GPR81 | 224131_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.8605
p=0.3649
p=0.2363
p=0.8036
p=0.0585
p=0.8639
p=0.1889
p=0.0485
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

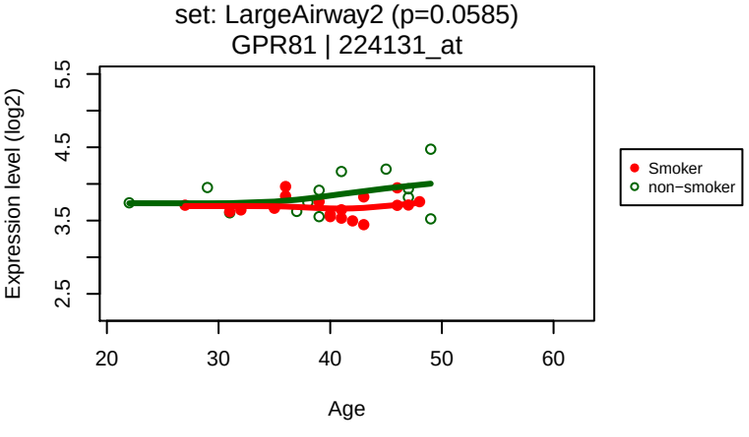

Gene expression ratio (smokers/nonsmokers)
gene: GPR81 | 224131_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8605
p=0.3649
p=0.2363
p=0.8036
p=0.0585
p=0.8639
p=0.1889
p=0.0485
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

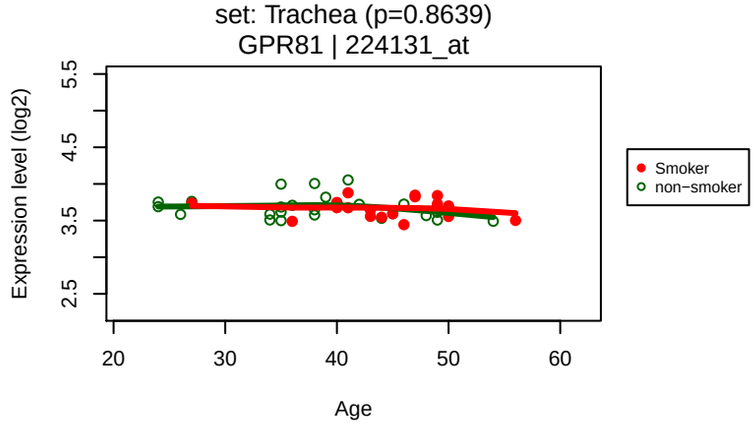

Gene expression ratio in Males
gene: GPR81 | 224131_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2528
p=0.4049
p=0.8388
p=0.0773
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: GPR81 | 224131_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6513
p=0.0433
p=0.806
p=0.2791
SmallAirway2
LargeAirway2
Trachea
Macrophage2

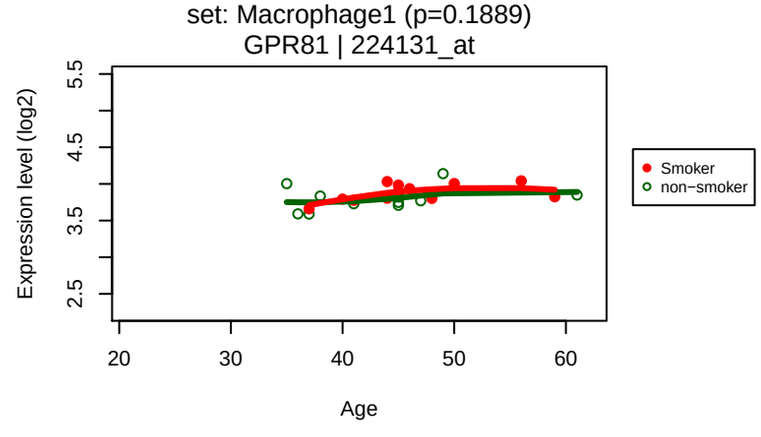

set: SmallAirway1 (p=0.8605)
GPR81 | 224131_at
Expression level (log2)
Age
Smoker
non−smoker

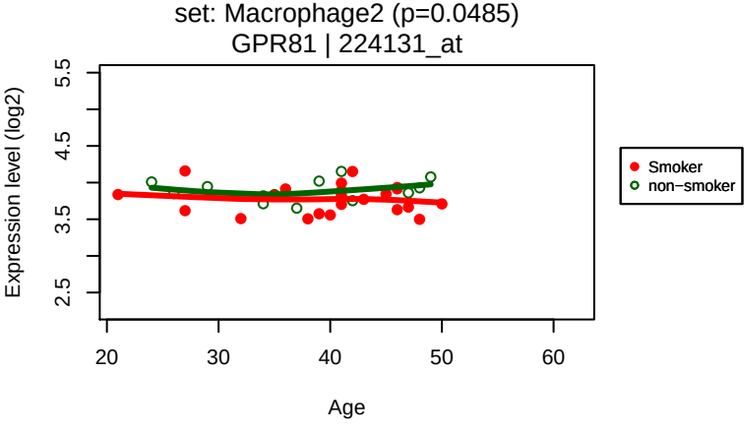

set: SmallAirway2 (p=0.3649)
GPR81 | 224131_at
Expression level (log2)
Age
Smoker
non−smoker

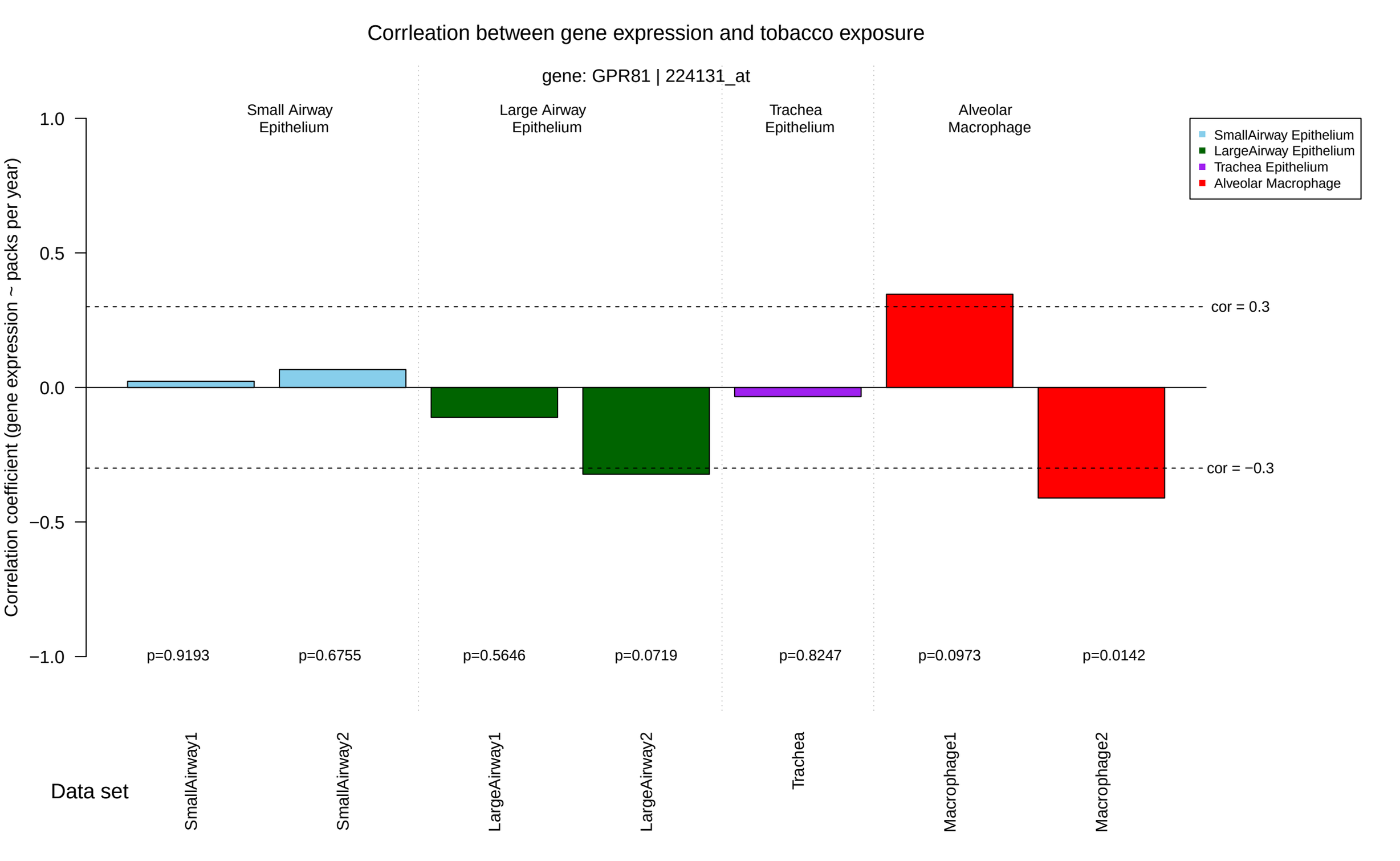

set: LargeAirway1 (p=0.8036)
GPR81 | 224131_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0585)
GPR81 | 224131_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.8639)
GPR81 | 224131_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.1889)
GPR81 | 224131_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0485)
GPR81 | 224131_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: GPR81 | 224131_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.9193
p=0.6755
p=0.5646
p=0.0719
p=0.8247
p=0.0973
p=0.0142
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2